

Conjecture 3 ([6]). For all $L, C \in \mathbb{N}$, there exists $C' \in \mathbb{N}$ such that if φ is an (L, C) -quasi-isometry from a graph G to a graph H , then there is an edge-weighting function $w: E(H) \rightarrow \mathbb{N}$ such that the same function φ is a $(1, C')$ -quasi-isometry from G to the weighted graph (H, w) .

This conjecture, if true, would imply that for any class of graphs $\mathcal{H}$ closed under contracting edges and taking subdivisions, if a graph G is (L, C) -quasi-isometric to a graph $H \in \mathcal{H}$, then G is $(1, C')$ -quasi-isometric to a graph $H' \in \mathcal{H}$ where C' depends only on L and C . This would be a remarkably powerful result if true.

In this paper, we disprove this conjecture by constructing explicit counterexamples demonstrating that multiplicative distortion for weighted graphs is, in fact, necessary.

Theorem 4. For every $C \in \mathbb{N}$, there exist graphs G and H and a $(2, 1)$ -quasi-isometry $\varphi: V(G) \rightarrow V(H)$ such that, for every edge weighting $w: E(H) \rightarrow \mathbb{R}^+$ of H , the map φ is not a $(1, C)$ -quasi-isometry from G to (H, w) .

Our proof for [Theorem 4](#) is based on orientated graphs H with large girth and chromatic number. We use the orientation of the edges to split each vertex into a new edge to define our graph G (see [Figure 1](#)). Since H can be obtained from G by contracting disjoint edges, the natural map φ between G and H is a $(2, 1)$ -quasi-isometry. Next, we assume that H is given an edge-weighting function $w: E(H) \rightarrow \mathbb{R}^+$. We then separate the light-weight edges from the heavy-weight edges and then find different types of long orientated paths within the graph. This allows us to find in H either a long path of light-weight edges that traverse many new edges, or a long path of heavy-weight edges that avoids the new edges. Since H has large girth, such paths are geodesic which allows us to contradict φ being a $(1, C)$ -quasi-isometry from G to (H, w) .

1.1 Preliminaries

Let G be a graph. The *girth* of G is the length of a shortest cycle in G . For $k \in \mathbb{N}$, a *proper k -colouring* of a G is a function $c: V(G) \rightarrow \{1, \dots, k\}$ such that $c(u) \neq c(v)$ whenever $uv \in E(G)$. The *chromatic number* $\chi(G)$ is the minimum $k \in \mathbb{N}$ for which G has a proper k -colouring.

Let (H, w) be a weighted graph. For a path $P = (v_0, v_1, \dots, v_n)$ in (H, w) , we say that the *length* of P is $\sum_{i=1}^n w(v_{i-1}v_i)$ and the *hop-length* of P is n . The path P is *geodesic* if it is a path of minimum length in (H, w) between v_0 and v_n .

An *oriented graph* $\vec{H}$ is a graph where each edge has a direction. The *chromatic number* of $\vec{H}$ is the chromatic number of H . Let $\vec{P}_n = (v_0, v_1, v_2, \dots, v_n)$ be an oriented path. We